

FOUNDATIONS & PRINCIPLES

01

Scale

vertical · horizontal · diagonal · scale cube (X-clone / Y-functional / Z-data) · sub-linear / linear / super-linear · coordination overhead

Performance

latency · throughput · bandwidth · Little's Law · latency budget · tail latency (P50/P95/P99/P999) · latency numbers

Reliability terms

availability (nines) · reliability · durability · MTBF · MTTR · SPOF · failure domain · blast radius · error budget

Failure modes

crash · omission · Byzantine · timing · gray failure · partial failure · cascading · metastable

Consistency

linearizable · sequential · causal · eventual · read-your-writes · monotonic read/write · session · tunable / quorum

Theorems

CAP · PACELC · harvest vs yield · FLP impossibility · end-to-end argument · Amdahl / Universal Scalability Law

Estimation

QPS · DAU/MAU · peak multiplier · read:write ratio · storage growth/retention · capacity planning · unit economics

NETWORKING & COMMS

02

Network

TCP / UDP · DNS (TTL, anycast) · HTTP/1.1·2·3 (QUIC) · TLS/mTLS · BGP · head-of-line blocking · split-brain

Sync / RPC

REST · GraphQL · gRPC + protobuf · Thrift / Avro · connection pooling · multiplexing · deadline propagation

Async / messaging

queue (P2P) · pub-sub · log/stream (Kafka, Pulsar, Kinesis) · at-most/least/exactly-once · ordering · DLQ · backpressure · dedup

Event-driven

event vs command vs query · notification · event-carried state · sourcing · choreography vs orchestration · schema registry · replay

API surface

gateway · BFF · composition/aggregation · versioning · pagination (offset/cursor/keyset) · webhooks · SSE · WebSocket · idempotency keys · OpenAPI

DATA MANAGEMENT

03

Storage engines

B-tree / B+tree · LSM-tree + SSTable · WAL · bloom filter · skip list · inverted index · row vs columnar · compaction

Database types

relational · document · key-value · wide-column · graph · time-series · vector/ANN · search · NewSQL · polyglot persistence

Partitioning

hash / range / directory / geo · consistent hashing · virtual nodes · partition key · hot partition / skew · scatter-gather · resharding

Replication

single / multi-leader · leaderless quorum ($R+W>N$) · sync / async / semi-sync · lag · LWW · CRDT · version vectors · Merkle anti-entropy · chain replication

Caching

cache-aside · read/write-through · write-behind · refresh-ahead · LRU/LFU/ARC/TTL · invalidation · stampede / penetration / avalanche · negative cache · L1+L2+CDN

Indexing

primary/secondary · covering · composite · partial · GSI/LSI · geospatial (R-tree, geohash, S2, H3) · bitmap · materialized view · write amplification

Modeling

normalization vs denormalization · CQRS · read model · outbox · CDC · schema evolution

COMPUTE & PROCESSING

04

Scheduling

cron / time-trigger · event-trigger · distributed cron (leader/lease) · job queue · priority queue · DAG orchestration · fan-out/fan-in · zombie reaping · misfire/overlap

Timers

timer wheel · hierarchical wheel · durable timer / delay queue · lease/TTL expiry · debounce / throttle · heartbeat

Batch

MapReduce · ETL / ELT · data lake · sort-merge join · checkpoint/restart · backfill · micro-batch

Stream

event vs processing time · windowing (tumbling/sliding/session) · watermark · late data · stateful + changelog · stream-table duality · exactly-once · Lambda vs Kappa

Aggregation

pre-aggregation / rollup · HyperLogLog · Count-Min sketch · t-digest/DDSketch · OLAP cube (Druid/Pinot) · sharded counter · leaderboard / top-K · eventual projection

RESOURCE & TRAFFIC

05

Rate limiting

token bucket · leaky bucket · fixed window · sliding-window log/counter · distributed (Redis/gossip/local-share/cell) · cost-based · 429 + Retry-After

Allocation & fairness

quota (hard/soft/burst) · fair queuing / WFQ · max-min fairness · DRF · priority + preemption · admission control · reservation · work-stealing · bin packing · noisy neighbor

Load balancing

L4 vs L7 · DNS/GSLB · round-robin / weighted · least-conn / least-time · consistent-hash · P2C (power of two) · health checks · connection draining · subsetting

Auto-scaling

reactive / predictive / scheduled · target tracking · queue-depth metric · cooldown / flapping · scale-to-zero / cold start · HPA/VPA/cluster autoscaler

RELIABILITY & RESILIENCE

06

Fault tolerance

retry + exp backoff + jitter · retry budget · circuit breaker (closed/open/half) · bulkhead · timeout / deadline · fallback / degrade · hedged requests · load shedding

Dist. transactions

2PC / 3PC · saga (choreo/orchestrated) · compensation · outbox · TCC · reservation · idempotency key · optimistic (ETag/version) vs pessimistic lock

Consensus & coord.

leader election (bully/ring) · Paxos / Multi-Paxos · Raft · ZAB · lease · distributed lock (ZK/etcd/Redlock) · fencing token · epoch/generation

Disaster recovery

RPO / RTO · backup (full/incr/diff) · PITR · pilot-light / warm / hot standby · chaos engineering · game day · immutable backup

ARCHITECTURE STYLES

07

Decomposition

monolith · modular monolith · microservices · service mesh · strangler fig · anti-corruption layer · DDD · bounded context

Layering

3-tier / N-tier · hexagonal (ports/adapters) · clean architecture · sidecar · ambassador · adapter

Isolation

cell-based · swimlane · shuffle sharding · bulkhead · blast-radius reduction · cell routing

Global

multi-AZ · multi-region active-passive / active-active · follow-the-sun · data residency/sovereignty · latency-based routing · region evacuation

Multi-tenancy

shared-everything · schema/DB-per-tenant · shared-nothing · tiered tenancy · tenant isolation · metering / cost attribution

IDENTITY & SECURITY

08

AuthN

session vs token (JWT/opaque) · OAuth2 (auth-code/PKCE/client-cred) · OIDC · SSO/SAML · MFA · mTLS · WebAuthn/passwordless · token rotation/revocation

AuthZ

RBAC · ABAC · ReBAC (Zanzibar) · policy engine (OPA/Cedar) · permission caching · token exchange · audit trail

Data security

encryption at-rest / in-transit · envelope encryption · KMS / HSM / Vault · tokenization / masking · secrets mgmt · zero-trust

OBSERVABILITY & OPS

09

Logging

structured logs · aggregation (ELK/Loki) · correlation ID · sampling · audit log · retention/rotation

Metrics

4 golden signals · RED · USE · counter/gauge/histogram · push vs pull · cardinality · SLI/SLO/SLA · burn-rate alert

Tracing

trace context (W3C/B3) · spans · head/tail sampling · OpenTelemetry · async-boundary tracing

Deploy / release

blue-green · canary · rolling · feature flags · dark launch · A/B · traffic shifting · rollback · expand-contract migration · IaC / immutable infra

SPECIALIZED SYSTEMS

10

Edge / CDN

PoP / origin shield · cache-control (s-maxage, SWR) · purge / tag invalidation · edge compute · adaptive bitrate (HLS/DASH) · ESI

Search

inverted index · BM25 / TF-IDF / LTR · tokenization/stemming · trie autocomplete · faceted · NRT indexing · vector/semantic + ANN

Notification

fan-out · push (APNS/FCM) · email pipeline (reputation/throttle) · preferences/channels · dedup/collapse · presence

Unique IDs

UUID v4 / v7 · Snowflake · ULID · DB sequence · ticket server · clock skew · URL shortener

Object storage

multipart / resumable upload · content-addressable · dedup · pre-signed URL · tiering (hot/warm/cold/archive) · lifecycle policy

Geo / location

geohash · S2 / H3 · quadtree · R-tree · proximity search · geofencing · map tiles

Pipelines

batch / streaming / Lambda · CDC (Debezium) · schema registry · data quality gate · orchestration (Airflow/Dagster) · lineage